

PLANNER DECISION VIEW

Release the polished-greedy plan

Exact recourse completed · independent validation passed · quality escalation available

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Decision summary

Planning metric	Default routing	Recommended plan	Change
Objective capture	74.96%	76.83%	+1.87 pp
Case fill	78.29%	79.91%	+1.62 pp
Reassigned orders	0	13	+13
Runtime	0.24 s	2.05 s	+1.81 s
Validation violations	0	0	No change

Objective capture is fulfilled value minus penalties and shipping, divided by total requested merchandise value. **Case fill** is fulfilled cases divided by requested cases. **pp** means percentage points.

Planner actions

1. Review the 13 reassignments for lane authorization and service rationale.
2. Confirm calendar exceptions and capacity events after the data snapshot.
3. Release only while the certification panel remains green.
4. Re-optimize after a material inventory change; trigger exact LNS when conflicts or penalty exposure are material.

What the solver certifies

- one assignment outcome per order and cohesive routing for each load;
- exact demand balance, nonnegative integral quantities, and eligible DC/date choices;
- protected cumulative ATP, documented capacity, and diversion-improvement compliance;
- independent objective recomputation and rejection of invalid or non-improving moves.

Escalation guide

Signal	Action
Routine window; plan passes review	Release polished-greedy plan
Coupled inventory/dock conflict, severe scarcity, or high penalty exposure	Run exact LNS with a fixed budget
Small disputed subset needs a bound	Run full MILP and inspect incumbent plus gap
Research proposal requested	Use an approved bounded/generated neighborhood; retain exact recourse and validation

Evidence boundary

All 513 returned plans in the final 516-row study pass validation; three frozen-routing controls are proven infeasible after 55% inventory reduction. At 70% reduction, adaptive methods remain feasible and exact LNS retains the strongest tested frontier. The study uses one supplied planning snapshot, so DC authorization, calendars, throughput maxima, customer rules, and commercial approvals still require owner confirmation.

Recommended decision: approve the polished-greedy plan subject to planner review, retain exact LNS as the quality escalation, and keep sampler-assisted runs behind the same certification gate.